

AlphaFold predictions on whole genomes at a glance: a coherent view on packing properties, pLDDT values, and disordered regions

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Abstract

For model organisms, AlphaFold predictions show that 30% to 40% of amino acids have a (very) low pLDDT confidence score. This observation, combined with the method’s high complexity, commands a systematic analysis of AlphaFold predictions on whole genomes.

Consequently, using whole-genome predictions, we provide a coherent analysis on packing properties, pLDDT values, and their relationship with intrinsically disordered regions (IDRs). Our contributions are of two kinds. First, we introduce simple and explainable geometrical and topological statistics characterizing predictions. Second, we investigate four key biophysical and biological questions: (i) the clustering of AlphaFold predictions on whole genomes, (ii) the identification of high/low quality predicted domains, (iii) false positive/negative AlphaFold predictions with respect to IDRs, and (iv) the fragmentation of the polypeptide chain in terms of pLDDT values.

Altogether, our analysis provide novel insights into AlphaFold predictions across whole genomes, further enhancing the confidence assessment of the models.

Keywords: AlphaFold, AlphaFold-DB, Model Quality Assessment, Topological persistence, Intrinsically Disordered Regions.

1 Introduction

1.1 AlphaFold: complexity and predictions

AlphaFold’s ability to reliably predict protein structures from amino acid sequences has ushered in a new era in structural biology [1]. In recognition of this achievement, its leaders D. Hassabis and J. Jumper were jointly awarded the 2024 Nobel Prize in Chemistry. Each AlphaFold prediction is provided as a PDB file, with individual amino acids assigned a confidence score known as pLDDT (predicted Local Distance Difference Test), related to the Local Distance Difference Test [2]. The AlphaFold team defined four confidence categories based on pLDDT values: very high ($0.9 \leq \text{pLDDT}$, dark blue), high ($0.7 \leq \text{pLDDT} < 0.9$, cyan), low ($0.5 \leq \text{pLDDT} < 0.7$, yellow), and very low ($\text{pLDDT} < 0.5$, orange).

On the experimental side, the Protein Data Bank currently holds approximately 230K experimentally determined structures, in contrast to the $\sim 200M$ protein sequences recorded in UniProtKB. This stark imbalance highlights the critical role of structure prediction in enabling genome-wide proteomic analysis (SWISS-MODEL Repository [3]). To address this, AlphaFold has been applied at scale, with predictions now available in the AlphaFold Protein Structure Database database—AlphaFold-DB, [4] and <https://alphafold.ebi.ac.uk>. A comprehensive analysis of this database reveals that for many model organisms, 30% to 40% of amino acids in predicted structures fall within the low or very low pLDDT confidence categories (Table 1, Section S6).

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From the methods’ perspective, AlphaFold uses two main modules, known as the **Evoformer** and the **structure module**. The overall algorithm is very complex, as evidenced by the ablation study which compares 10 variants to assess the merits of the main building blocks [1, SI]. The overall architecture heavily relies on transformers [5]. Beyond unsupervised approaches such as direct coupling analysis [6] or MSA transformers [7], the supervised inference of contacts by transformers within the **EvoFormer** is a key asset of AlphaFold [8]. In the context of the bias-variance trade-off in machine learning in general [9], and of the double descent phenomenon for large deep models [10], an important point is the balance between the number of parameters in AlphaFold and the pieces of information used to train it. The number of parameters has been estimated circa 100 millions. On the other hand, the number of non redundant sequences used in the self-distillation procedure is 350k [1, Section 1.3, SI], which, assuming a median size of 500 amino acids, results in about $\sim 1.75 \cdot 10^8$ amino acids. (Distillation is the process of transferring knowledge from a teacher to a student [11], a simplifying process bearing similarities to coreset calculations in a more algorithmic setting learning [12].)

AlphaFold predictions: structural analysis. The structural features within AlphaFold predictions have been examined using the **Geometricus** algorithm[13], which represents protein structures through a series of steps: (i) a decomposition of the protein into structural fragments (based on k -mers or spherical neighborhoods), (ii) a discrete encoding of invariants (moments) of these fragments into shape-mers, and (iii) a counting step of shape-mers yielding an embedding of the protein. By clustering these embeddings – using non-negative matrix factorization – across 21 predicted proteomes from AlphaFold-DB, 250 structural clusters were identified, including 20 major clusters or superfamilies [14].

AlphaFold predictions and intrinsic disorder. The pLDDT metric has been thoroughly analyzed, revealing its effectiveness as an indicator of result robustness. It has been observed that the low-pLDDT regions in AlphaFold predictions correlate with observed or predicted intrinsically disordered regions (IDRs) [15]. Indeed AlphaFold outperforms state-of-the-art methods such as IUPred2 on disordered region detection on a benchmark of DisProt targets and PDB controls [14]. Also, high mean pLDDT values are associated with minimal structural differences between AlphaFold and trRosetta models [14]. However, as indicated by a lack of association with B-factors, pLDDT does not appear to correlate with flexibility [16]. Although pLDDT can predict structural disorder, alternative estimators using AlphaFold-predicted structures, such as the relative solvent accessibility (RSA), provide an even more accurate measure of disorder [17].

1.2 Contributions

The significant percentage of amino acids with a (very) low pLDDT and the AlphaFold complexity call for further investigations to better understand the quality of predictions (Table 1). More specifically, using whole-genome predictions, our goal is to provide a coherent perspective on packing properties, pLDDT values, and their relationship with intrinsically disordered regions or proteins (IDPs/IDRs). Our contributions are of two kinds: first, we introduce two geometric/topological analysis methods; second, using these methods, we investigate four key biological questions.

Methods. Topological data analysis provides a set of tools and concepts to extract stable features from high-dimensional data [18, 19]. To understand properties of AlphaFold predictions, we use the concept of *filtration*, namely a progressive encoding of a structure as a sequence of nested shapes starting with the empty set and ending with the entire structure, monitored by a real-valued parameter. In practice, we use two filtrations: the first one is associated with the number of neighbors of a C_α carbon—a local packing measure; the second one is based on the pLDDT value provided by AlphaFold. Distributions associated to these filtrations are used to compare structures using optimal transport theory [20, 21], and for pLDDT based analysis, to assess structures with respect to a null (random) model.

These tools are used to study four biological questions.

Q1. Contacts, packing, and whole genome predictions. Given the prominent role played by contacts in the EvoFormer, our first analysis focuses on packing properties of predictions. An embedding of AlphaFold predictions based on shape-mers has been obtained using dimensionality reduction (t-SNE, [14, Fig. 2]), but the corresponding non linear coordinates are not easily exploited to study correlations with other properties.

In search of an embedding performing at once dimensionality reduction and clustering in the context of packing properties, we introduce the *arity*, an encoding based on percentiles of the distribution representing the number of neighbors of each amino acid. Such a distribution can be summarized by a small number of integers. Using two numbers yields the *arity map*, a 2D natural low dimensional embedding of all structures, which we use to investigate the quality of domains predicted by AlphaFold and the relationship with IDPs/IDRs.

Q2. Predicted domains and their quality. The aforementioned complexity of AlphaFold and the commensurate information size raises the question of potential biases in the model. To address this, we examine the locations of predicted low-quality domains. To understand our approach, it is important to recall that protein domain classification is a cornerstone of protein science [22, 23, 24]. A particularly noteworthy effort is the recently published ECOD structural domain classification of the entire human proteome [25], which leveraged AlphaFold predictions to enrich its classification with several new structural domains categorized by confidence scores as high or low. A domain is classified as low-confidence if it appears globular by its predicted aligned error (PAE) description but lacks a strong homologous hit (via a HMM calculation) within the ECOD reference [25] and prodata.swmed.edu/ecod/.

We investigate whether the training of AlphaFold introduces biases in the prediction of low-quality domains by analyzing their distribution across the arity map. Notably, we identify a specific region of the arity map that is disproportionately populated by such domains.

Q3. Predictions and intrinsically disordered proteins/regions. The per-residue pLDDT scores on the AlphaFold-DB human proteome prediction yields a bimodal distribution where the two peaks identify ordered and disordered regions [26]. Nonetheless, there are exceptions: conditionally folded IDRs are usually predicted as stably folded regions, and studies seems to imply that 10% of the low-pLDDT regions are not related to disorder [26].

To complement these findings, we show that although most of the AlphaFold disorder predictions are supported by experimental evidence, predictions with certain structural characteristics (a specific region in the arity map) are much richer in false positive IDR detection than others.

Q4. pLDDT values and fragmentation of AlphaFold reconstructions. The pLDDT value is a key statistics to assess AlphaFold predictions, but its coherence along the primary structure has not been studied. We provide this analysis via a multiscale characterization of stretches along the sequence with coherent pLDDT values. This analysis relies on the stability of connected components of amino acids (a.a.) along the protein sequence, upon building a suitable filtration which consists of inserting the a.a. by decreasing pLDDT values.

2 Methods

2.1 Packing analysis

Arity and arity signature. A natural parameter to identify compact regions in proteins in the number of neighbors of a C_α carbon. Let the *arity* of a C_α carbon be the number of neighboring C_α s within a distance range r . As classically done, we take $r = 10\text{\AA}$ to account for non-covalent contacts. We note in passing that such neighbors are sufficient to robustly infer protein domains using a direct application of spectral clustering[27]. Assuming the polypeptide chain has n amino acids, define:

- $L_a = \{a_n, \dots, a_n\}$: arities of the n C_α carbons;
- $L = [A_1, \dots, A_m]$: unique arities sorted by increasing value;

The number of C_α with a given arity is used to define the histogram H_{C_α} . We compare two normalized such histograms (unit cumulated sum) using the mass transport solver from [28] implemented in the *Python Optimal Transport* package [29].

To simplify the presentation provided the H_{C_α} and the associated CDF_{C_α} , we define the following signature:

Definition. 1 (*Arity signature $Sig_{C_\alpha}(P)$ of a polypeptide chain P .*) Consider the cumulative distribution functions associated with the number of C_α carbons, that is CDF_{C_α} . Consider a list $P_K = \{p_1, \dots, p_K\}$ of increasing percentiles. For $p_k \in P_K$, let $\text{arity}_{C_\alpha}(p_k)$ be the smallest arity such that the CDF is larger than p_k , that is

$$\text{arity}_{C_\alpha}(p_k) = CDF_{C_\alpha}^{-1}(p_k) \stackrel{\text{Def}}{=} \arg \min_j CDF_{C_\alpha}(A_j) \geq p_k. \quad (1)$$

The C_α signature is the increasing sequence $Sig_{C_\alpha}(P) = \{\text{arity}_{C_\alpha}(p_1), \dots, \text{arity}_{C_\alpha}(p_K)\}$.

Using two quantiles $\{q_1 = 0.25, q_2 = 0.75\}$, the arity signature refers to the two arity values required to gather 25% and 75% percent of the number of amino acids in the chain (Fig. 1 for prototypical folds).

For a collection of structures, typically all predictions in a genome, we define:

Definition. 2 Given two quantiles $q_1 (= 0.25)$ and $q_2 (= 0.75)$, the arity map is the map whose x and y axis are the arities at q_1 and q_2 . A bin/cell of the map hosts all structures with a prescribed arity signature.

Upon processing the C_α carbons by increasing arity, we are also interested in the number of connected components of the primary structure defined by all amino acids / C_α s whose arity is less than a given threshold A_i . We now formalize this idea using topological persistence.

2.2 Persistence based analysis on the primary structure

Filtrations and persistence diagrams. We aim to evaluate the *coherence*, with respect to a parameter generically denoted as u , of the n amino acids of a protein sequence in an AlphaFold prediction. Practically, u will be the arity or the pLDDT, so that we expose the method for a general value u .

We consider the polypeptide chain as a path graph with n vertices and $n - 1$ edges. From a topological data analysis perspective [18], we use a filtration $\mathcal{G}_u$ encoding a sequence of nested subgraphs of this path graph. Here, the calligraphic letter $\mathcal{G}$ indicates that the filtration is applied to the path graph, and u is the value carried by each amino acid. Upon inserting a.a. by increasing u value, we connect the a.a. being processed to its neighbors along the sequence if already inserted—Algorithm 1. The filtration starts with the empty graph and ends with the complete path graph.

As u varies, we track the number of connected components (c.c.) in the graph $\mathcal{G}_u$ using the Union-Find algorithm [30]. This results in a function $\nu = N_{cc}(u)$ representing the number of connected components at each value of u , with ν in the interval $[1, \lceil n/2 \rceil]$.

The *persistence* of a connected component in $\mathcal{G}_u$ is the time elapsed between its birth and the moment when it merges with a c.c. that was created earlier. The *persistence diagram* (PD) is the diagram whose x -axis (resp. y -axis) is the birth (resp. death) date [18]. The points of the PD are denoted $P = \{c_1, \dots, c_m\}$, and referred to as *critical points*. The persistence of point c_i reads as $p_{\mathcal{G}}(c_i) = \text{death}_{\mathcal{G}}(c_i) - \text{birth}_{\mathcal{G}}(c_i)$. We also denote m' the number of critical points whose persistence is positive—i.e. critical points away from the diagonal, and define the fraction of positive critical points as $f_{cp}^+ = m'/m$.

Assuming that the persistences belong to a discrete set $\mathcal{P}$, we denote $\mathbb{P}[\cdot]$ the probability mass function $\mathbb{P} : \mathcal{P} \rightarrow [0, 1]$. To assess the diversity of persistence values, we consider the normalized entropy of this probability distribution.

Finally, we consider the *salient / persistent* local maxima of the function $N_{cc}(u)$. We identify such maxima using persistence again on the filtration $\mathcal{H}_\nu$ associated with super-level sets $N_{cc}^{-1}([\nu, \lceil n/2 \rceil])$ of the function $N_{cc}(u)$. (The calligraphic letter $\mathcal{H}$ indicates that the filtration is on the height function $N_{cc}(u)$.) The persistence $\mathbf{p}_{\mathcal{H}}(c_i) = \text{death}_{\mathcal{H}}(c_i) - \text{birth}_{\mathcal{H}}(c_i)$ of a local maximum of $N_{cc}(u)$ is the elevation drop leading to a saddle point also connected to a more elevated local maximum. We simplify the function $N_{cc}(u)$ using the persistence simplification based on the Morse-Smale-Witten chain complex [31]. In particular, we denote $\text{PLM}(t_\nu)$ the number of persistent local maxima of $N_{cc}(u)$ at persistence threshold t_ν . In practice, we will use a relative threshold $t_p \in (0, 1)$ to define $t_\nu = n * t_p$ – with n the number of amino acids.

We summarize the previous discussion with the following:

Definition. 3 Using the filtration $\mathcal{G}_u$ coding subsets of the path graph connecting the n amino acids of a polypeptide chain, we define:

- The fraction of positive critical points: $f_{cp}^+ = m'/n$.
- The mean persistence : $\bar{\mathbf{p}} = \sum_{i=1}^m \mathbf{p}_{\mathcal{G}}(c_i)/m$,
- The normalized persistence entropy :

$$H_p = - \sum_{\mathbf{p}_i \in \mathcal{P}} \mathbb{P}[\mathbf{p}_i] \log \mathbb{P}[\mathbf{p}_i] / \log |\mathcal{P}|. \quad (2)$$

Using the filtration $\mathcal{H}_\nu$ consisting of super-level sets of the function $N_{cc}(u)$, we define:

- The maximum number of connected components observed: $N_{cc}^{max} = \max_u N_{cc}(u)$
- The number of persistent local maxima of the function $N_{cc}(u)$ at persistence threshold t_ν : $\text{PLM}(t_\nu)$.

We are now in position to define the two filtrations of interest.

Arity based filtration $\mathcal{G}_{\text{arity}}$. Using the arity as filtration parameter yields the filtration $\mathcal{G}_{\text{arity}}$, whose connected components are maintained using Algorithm 2—a simple variant of Algorithm 1 handling all C_α carbons with a given arity in a batch.

As an illustration, we consider the evolution of connected components for prototypical folds (Fig. 1).

pLDDT based filtration. The filtration $\mathcal{G}_{\text{pLDDT}}$ uses $-\text{pLDDT}$ as parameter u . We indeed process $-\text{pLDDT}$ values in increasing order, to handle high quality regions first. Assuming that all pLDDT values are different, the PD therefore contains exactly $n-1$ critical points corresponding to the $n-1$ edges connecting the n residues. A number of these have a null persistence and correspond to *accretion*—a C_α carbon inserted in the Union-Find data structure, and merged immediately to a neighbor on the left or the right. These account for the definition of the fraction $f_{cp}^+ = m'/(n-1)$ (Def. 3).

Example 1 Consider first a fictitious protein with random pLDDT values in $[0, 100]$. The insertion of individual a.a. results in a well behaved curve peaking at the median pLDDT value, and whose maximum is conjectured to be $1/4$ of the number of a.a. (Fig. 2, Fig. S1; conjecture 1). The persistence diagram also exhibits a characteristic triangular shape with a concentration of critical points near its apex, corresponding to the death of connected components upon moving forward to the full path graph in $\mathcal{G}_{\text{pLDDT}}$. Simulations also give the following approximate values: $n = 100$: $f_{cp}^+ \sim 0.33$, $H_p \sim 0.36$; $n = 1000$: $f_{cp}^+ \sim 0.33$, $H_p \sim 0.45$; $n = 10,000$: $f_{cp}^+ \sim 0.33$, $H_p \sim 0.47$.

Consider next the following fictitious protein: (i) a first group of a.a. corresponding to an unstructured region on the N-ter side, with low confidence (say $\text{pLDDT} \in [0, 49]$; and (ii) a second group of a.a. corresponding to a structured region on the C-ter side, with high confidence (say $\text{pLDDT} \in [50, 100]$). Along the insertion, one observes the formation of the former and the latter regions (Fig. 2).

3 Results

3.1 Q1. Contacts, packing, and whole genome predictions

Charting whole genomes with the arity map. The arity is instrumental at distinguishing ordered, disordered, and mixed proteins. While proteins with classical α and β folds exhibit an arity in the range 15-20, a mode in the interval 5-7 is observed for unstructured regions (Fig. 1, Fig. 3). See also Sec. S1.2 for numerous additional examples.

We process all proteins of a given genome and build the *arity map* (H. Sapiens, Fig. 4(A)). The interpretation of this map provides a whole genome analysis at a glance. The x (resp. y) axis $\text{arity}_{C_\alpha}(q_1)$ (resp. $\text{arity}_{C_\alpha}(q_2)$) is geared towards loosely (resp. tightly) packed regions. For a given cell of this map, the distance to the diagonal $y = x$ encodes the coherence of the overall packing of the region of the protein selected by the quantiles q_1 and q_2 . For example, for bins/cells on the diagonal, a unique arity values suffices to move from a fraction of amino acids $< q_1$ to a fraction $> q_2$, stressing the coherence of the overall 3D structure. This is never the case for native protein structures, which always present different quantiles q_1 and q_2 (Fig. S12, in red) due to their interchange between rigid and flexible regions. But such a collapse is observed on AlphaFold predictions (Fig. 4, AF-Q8IWJ2-F1).

To further analyse proteins in a cell of the arity map, we perform a hierarchical clustering of arity histograms using the Earth Mover distance as metric [20, 28]. As expected, the clustering unveils the finer structure of those proteins (Fig. S10(B,C,D)).

We produced the arity map for all models genomes from AlphaFold-DB (H. Sapiens, Fig. 4; other species, Sec. S5). In passing, we note that arity and pLDDT strongly correlate (Fig. S11), confirming that the arity is also a good proxy for prediction robustness. Yet, pLDDT is not able to show the finer details of the structural repartition that emerges from the arity map.

These maps also appear to be bimodal. The most pronounced mode corresponds to well folded proteins, in the arity map region $[14, 18] \times [20, 25]$. The second mode corresponds to the region $[4, 8] \times [10, 15]$, with the particularly populated cell 7×13 . These modes are in general separated by low density regions. The inspection of genomes beyond the model ones yields different patterns, though. For example, *P. Falciparum* exhibits a continuum between the previous two modes, which is consistent with the overall low quality of predictions noted above.

We further investigate the hot spot at 7×13 of the human genome (Fig. S10). Contrary to the rest of the $[4, 8] \times [10, 15]$ region, this hot spot does not find any counterpart in the arity plot of the human proteins of known structure (Fig. S12, in red) – and is not present either in the plot of experimentally-confirmed disordered proteins (Fig. 6, discussed later). These AlphaFold-DB predictions therefore represent disordered structures or inaccurate predictions.

The connexion with IDRs/IDPs is the object of our Question 3.

Arity based filtration. The variability of arities coding the compactness of the atomic environments around C_α s allows incrementally *building* the protein by inserting the a.a. by order of increasing arity and connecting each a.a. to its predecessor (N-ter) and successor (C-ter) if already inserted (Section 2.1, Algorithm 2).

The case of well folded proteins follows this pattern as focusing on salient extrema of the function $N_{cc}(\text{arity})$ yields a unimodal arity curve with a salient extremum in the range 15-20 (Fig. 3(Left column, second row, blue curve)). The same holds for unstructured proteins, excepts that the local extremum of $N_{cc}(\text{arity})$ is attained at values $\sim 5 - 7$ (Fig. 3(Middle column, second row, blue curve)). The signature is more informative for proteins having both high and low quality regions (Fig. 3(Right column, second row, blue curve)). For such proteins, the curve of connected components is typically bimodal, since one successively forms the regions with low and high arity.

Concluding, the ability to detect the persistent local extrema on these curves gives a direct indication on the structural elements encountered in the predictions.

3.2 Q2. Predicted domains and their quality

We investigate potential learning biases in AlphaFold predictions at the sequence and structure levels.

At the sequence level, counting the number of homologous structures in UniProtKB for each arity signature exhibits a clear trend between the arity signature and the average number of homologous sequences (Section S1.2, Fig. S13(A)). However, mapping these homologs to their cluster representatives in the current release of Uniclust30 erases this bias (Fig. S13(B)), simply showing that structured proteins have more homologous sequences in nature. We also note that the quality of predictions in terms of average pLDDT shows a gradient towards high arity values (Fig. S12). Since both the AlphaFold direct and distillation training algorithms construct MSAs of non-redundant sequences with sequence identity $< 30\%$, we conclude that a large and diverse input MSA is necessary but not sufficient for guaranteeing high-quality predictions.

We proceed with the investigation of potential biases at the structural level using the recently published ECOD structural domain classification of the entire human proteome [25]. Using AlphaFold-DB predictions, ECOD could enrich its classification with several new structural domains. At the level of H-groups in ECOD hierarchy, each identifier refers to a cluster of sequence-homologous, same-topology domains in tight relation with the SCOP superfamily level. For each arity signature, we thus collect the unique ECOD H-group identifiers and the corresponding folds. The resulting 2D-histogram shows a clear correlation with the protein count and the number of ECOD folds (Fig. S14(A), upper triangle for count and lower triangle for ECOD folds). In order to remove the correlation and study the residual signal, we consider the signed difference of the two rescaled histograms. (The rescaling ensures that the two maxima are equal.) This difference map shows two main domains (Fig. S14(B), red and blue regions). On the one hand, the region of the main peak *i.e.* 13×20 until the 7×13 peak (excluded) shows an enrichment in ECOD domains. On the other hand, the region of the 7×13 peak towards the bottom left of the arity map (IDP region, see next Section) gets depleted.

ECOD also provides a set of low-confidence domains associated with low pLDDT and PAE (predicted aligned error) values [25], enabling the calculation of the enrichment relative to this dataset. Remarkably, the former *red* region gets split, the enrichment in low confidence domains being confined to the 7×13 signature as well as the arity signatures $[5, 15] \times [14, 20]$. This extended region contains structures richer in IDRs, but still having well-formed domains. Conversely, the $[5, 7] \times [5, 12]$ region is poor both in standard and low-confidence well-formed domains, hinting at a complete dominance of IDRs over structured regions. These observations call for a further inspection of IDPs/IDRs in the arity map.

3.3 Q3. Predictions and intrinsically disordered proteins/regions

Several recent works assume that in AlphaFold's predictions, regions with low pLDDT and/or not displaying secondary nor tertiary structure are to be considered intrinsically disordered rather than just the product of insufficient or unexploitable information. The tools developed in the previous sections can help assessing whether and when a correlation between unstructured regions in AlphaFold's prediction correlate with experimental evidences of disordered regions.

Analysis using DisProt. We first consider the DisProt database, which collects disordered regions from more than 1300 target protein sequences. Many of the benchmarked sequences correspond to proteins of known structure, present in the PDB. For this test, we focus on the 266 targets lacking structural data and that have a predicted structure on AlphaFold-DB. Consider all a.a. of all proteins in a given arity map cell. We compute the fraction of false positive of these – a.a. with a pLDDT $< 50\%$ but lacking a disorder annotation in DisProt. Notably, DisProt is known not to provide complete annotations, thus in principle AlphaFold could discover new unannotated IDRs on the DisProt targets. The simplest hypothesis consists of expecting an incompleteness proportional to the extent of the predicted disordered regions – regardless of other structural characteristics of AlphaFold predictions. However, while AlphaFold predicts a large number of amino acids as part of IDRs for the region of the arity map $\text{arity_}C_\alpha(0.25) < 7$, most of these are not annotated as such in DisProt (Fig. 6(A)). Predictions populating that part of the arity plot are thus likely to be of poor quality: this is not surprising, as the previous section details that proteins in this region often have a low number of homologous structures.

Analysis using AIUPred. Finally, we reconsider the 7×13 cell of the arity map of the human genome, hosting candidates for divergent AlphaFold predictions (Fig. S10). We ran the state-of-the-art software AIUPred [32] to predict IDRs on the whole human genome, and for each protein computed the number of residues characterized by $pLDDT \geq 0.5$ and a disorder score of > 0.5 according to AIUPred (Fig. 6(B)). The triangle delimited by $([6, 8], [6, 13], [12, 13])$ is enriched in predicted structures not clearly recognized by AlphaFold. This region is somehow complementary to the one highlighted by DisProt, and interestingly, the only neighborhood that both displays false positive and false negative IDR predictions is centered in 7×13 .

3.4 Q4. pLDDT values and fragmentation of AlphaFold reconstructions

We use the filtration $\mathcal{G}_{pLDDT}$ and the associated values (Def. 3, Fig. 3, Fig. 7) to characterize the coherence of pLDDT values along the backbone.

Specific proteins with different profiles. To appreciate the understanding yielded by our statistics, and in particular the persistence entropy H_p and the variation of the number of connected components $N_{cc}(pLDDT)$, let us consider various protein profiles, namely well structured, disordered, and mixed (Fig. 3). It appears that the function $N_{cc}(pLDDT)$ varies from unimodal (when considering persistent maxima) to multimodal. The former occurs for well structured or unstructured proteins (Fig. 3(Left, Middle)), and the latter for mixed cases (Fig. 3(Right)). For example, the *D. Melanogaster* protein AF-Q9VQS4-F1-model_v4 exhibits a clear bimodal profile with two persistent maxima (Fig. 3(Right)). On this example, the plateau at five connected components corresponds to five helices—four long ones plus a shorter one. The persistence diagram associated with the construction of the filtration also exhibit the dynamics of connected components (Fig. 3, fourth row). Using the same example, the PD of *D. Melanogaster* quantifies the dynamics of the high (resp. low) confidence region of the molecule on the left (resp. right) part of the diagram. We note that the apex of the triangles are not as populated as those of the null model (Fig. S1), showing a lesser randomness of pLDDT values, which is consistent with the lesser entropy H_p .

Let us now analyse these quantities in more detail at the genome scale.

Persistence entropy. At the genome scale, we observe a strong correlation between the three pairs of variables amidst f_{cp}^+ , H_p , $\bar{p}$ (Fig. 7, Fig. S15). The maxima of f_{cp}^+ are consistent with those of the null random model, while those of H_p are slightly below those of the same null model. (NB: average values of $\bar{p}$ are irrelevant for the null model.) For all organisms, the minimum H_p is observed for very regular structures for which all pLDDT values are within a small range (Fig. S16(Second row)). Instead, the maxima are observed for highly disordered structures whose a.a. span a wide range of pLDDT values (Fig. S16(Third row)). For such cases, the typical triangular shape in the PD is also observed. In considering the overall pLDDT statistics per organism (Table 1, SI Section S6), it appears that shifting the pLDDT values towards 100 results in a lesser fraction of positive critical points and in a smaller entropy (Fig. S15).

Maximum value N_{cc}^{max} of the function $N_{cc}(pLDDT)$. The maximum number of local maxima observed N_{cc}^{max} normalized by the protein size (Fig. S17) in general lower than that of the null model – conjectured to be equal to $1/4$, even though some cases reach or even surpass that bound. This is accounted for by two reasons. First, accretion is at play—see definition above. Second, pLDDT values come in batches, so that local insertions into *stretches* along the sequence prevent the maximum number of local maxima to rise. These batches are identified by the persistent local maxima discussed next.

Persistent local maxima PLM of the function $N_{cc}(pLDDT)$. We first note that the number of persistent local maxima is in general less than 10 (Fig. S15(Third column)). Also, the correlation between H_p (or f_{cp}^+) and PLM is quite low, which is expected. One can indeed have a significant number of persistent critical points localized in a narrow range of pLDDT values (Fig. S18). This phenomenon is well known to geographers defining peaks on mountains: two tall peaks can be separated by a low lying pass, but be

located at a short flying distance [33]. Nevertheless, as pointed out above, persistent local maxima identify coherent stretches along the sequence in terms of pLDDT values.

Model assessment and database queries. In the model assessment perspective, querying the encoding provided by the number of a.a., the persistence entropy H_p , the fraction of positive critical points f_{cp}^+ and the number of persistent local maxima PLM enables the identification of cases incurring a large fragmentation. Such cases exhibit a wide range of pLDDT values aggregated in coherent stretches along the sequence, yielding a large entropy and a (relatively) high number of persistent local maxima of the function $N_{cc}(pLDDT)$ (Fig. 8).

4 Discussion and outlook

AlphaFold has significantly advanced protein structure prediction, delivering remarkably high accuracy across a wide range of targets. A key feature of its predictions is the pLDDT score, which offers an estimate of local confidence for each amino acid of the predicted structure. This score is computed by AlphaFold’s structure module—a component that has since been widely adopted in other structure prediction tools. pLDDT has also been shown to anticorrelate with intrinsically disordered regions, surpassing the accuracy of dedicated state-of-the-art disorder predictors. Despite its impressive overall performance, AlphaFold is not without flaws: there have been consistent reports of clear prediction errors—cases where incorrect structures are assigned high average pLDDT scores [34, 35, 36]—yet identifying a shared cause behind these failures has remained elusive. Only recently have large-scale, first-principles analyses begun to emerge for protein language models (pLMs) [37].

In this context, our work introduces a novel representation of structural elements in proteins with two powerful descriptors at the amino acid level, which are formalized as topological filtrations.

The first filtration describes the 3D packing properties and the corresponding cumulative distribution counting the number of C_α carbons with a prescribed number of neighbors. This distribution facilitates the comparison of structures using optimal transportation, and the persistent components of the filtration can be used to compare structural motifs from different proteins. To simplify this representation, the distribution can be reduced to two integers, forming the *arity map*, which simultaneously performs clustering of coherent structures and dimensionality reduction. The assignment of structures to cells of the arity map charts the space of predictions from the structural standpoint, and provides an embedding of structures that enables the study of certain properties *as a function* of the arity. In particular, the coherence of observations for neighboring cells being unveils important trends in two directions. First, the study of domains yields the identification of a specific region of the arity map populated with low quality structures. The region is enriched in low-confidence domains recently published in the AlphaFold enriched version of the ECOD database. Most of these domains may be termed *hallucinations* since they may appear valid, but are incorrect and/or fabricated. Second, the investigation of IDRs/IDPs using the arity map also hints at another specific region with a high rate of false positive IDRs. While the performances of AlphaFold for IDP prediction have been praised, the notion of isolated structure for IDPs is iconoclastic since dynamics play a key role, and future work will likely address the generation of ensembles.

The second filtration uses pLDDT values along the sequence. This filtration and the associated statistics characterize cases with a wide range of pLDDT values aggregated in coherent stretches along the sequence, yielding a large entropy and a (relatively) high number of persistent local maxima of the function counting connected components. These analysis make it possible, in particular, to isolate regions of a model with a coherent quality (pLDDT) profile.

Overall, we believe that the statistics developed in this work complement those already delivered within AlphaFold-DB, and will contribute to understand limitations of AlphaFold regarding low quality domains and false positive IDRs.

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genome / pLDDT	50	70	90
AThaliana/all	0.204	0.315	0.550
CAlbicans/all	0.213	0.312	0.570
CElegans/all	0.202	0.323	0.597
DDiscoideum/all	0.288	0.423	0.681
DMelanogaster/all	0.281	0.387	0.623
DRerio/all	0.246	0.343	0.593
EColi/all	0.029	0.078	0.275
GMax/all	0.217	0.338	0.577
HSapiens/all	0.284	0.382	0.666
MJannaschii/all	0.035	0.084	0.264
MMusculus/all	0.255	0.352	0.597
OryzaSativa/all	0.257	0.408	0.623
RattusNorvegicus/all	0.253	0.351	0.596
SCerevisiae/all	0.213	0.314	0.582
SPombe/all	0.187	0.289	0.558
ZeaMays/all	0.260	0.394	0.635
SAureus/all	0.045	0.096	0.295
HPylori/all	0.053	0.123	0.347
MTuberculosis/all	0.069	0.133	0.322
Aeruginosa/all	0.036	0.086	0.278
PFalciparum/all	0.460	0.584	0.804

Table 1: **Cumulated distribution function for pLDDT values of all a.a. of all proteins in a genome.** See section **S6** for statistics on median and mean values. Top: model genomes. Bottom: five *global health* proteomes. Min and max values per column stressed in bold.

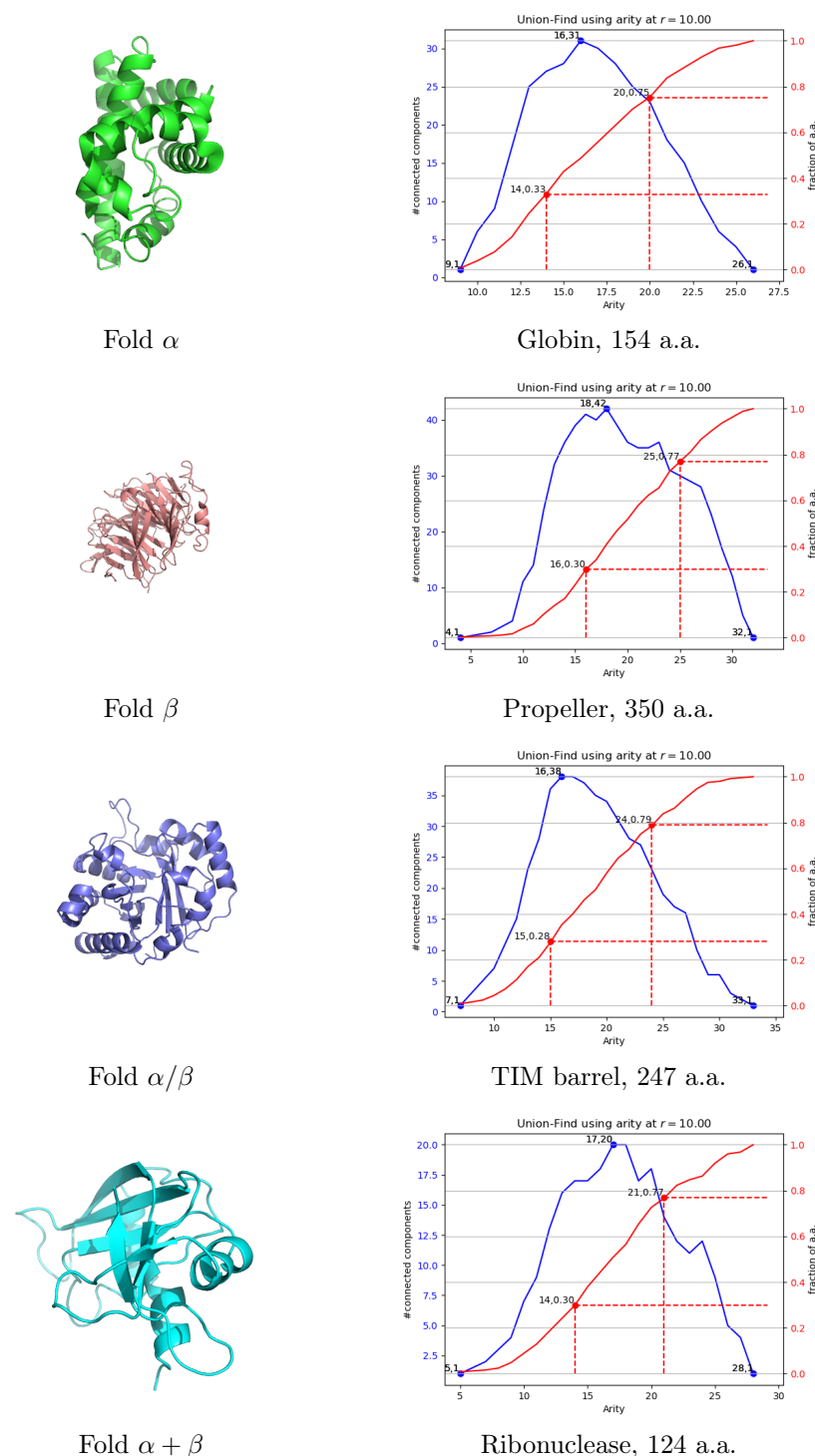


Figure 1: **Arity distribution and arity signature for prototypical folds.** Fold α : 101m-globin-alpha; 154 amino acids. Quantile-arity signature [(0.25, 14), (0.5, 17), (0.75, 20)] Fold β : 1erj-propeller-beta; 350 amino acids. Quantile-arity signature [(0.25, 16), (0.5, 20), (0.75, 25)] Fold α/β : 8tim-TIMbarrel-alphaSlashBeta; 247 amino acids. Quantile-arity signature [(0.25, 15), (0.5, 19), (0.75, 24)] Fold $\alpha + \beta$: 1a5p-ribonuclease-alphaPlusBeta; 124 amino acids. Quantile-arity signature [(0.25, 14), (0.5, 17), (0.75, 21)]

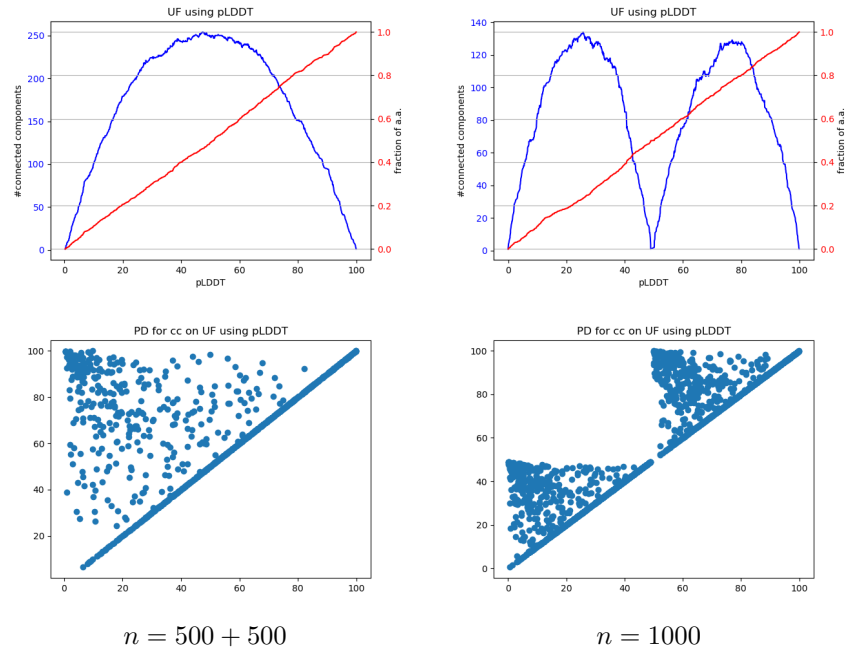


Figure 2: **Filtration $\mathcal{G}_{\text{pLDDT}}$ and Union-Find: evolution of the number of connected components $N_{\text{cc}}(\text{pLDDT})$ for fictitious proteins with one domain and two domains, respectively.** Left: the $n = 1000$ a.a. are assigned random pLDDT values in $[0, 100]$; Right: the first 500 a.a. are assigned random pLDDT values in $[0, 49]$, and the last 500 a.a. are assigned random pLDDT values in $[50, 100]$. **(Top row)** Blue curve: function $N_{\text{cc}}(\text{pLDDT})$; red curve: fraction of amino acids. **(Bottom row)** persistence diagrams.

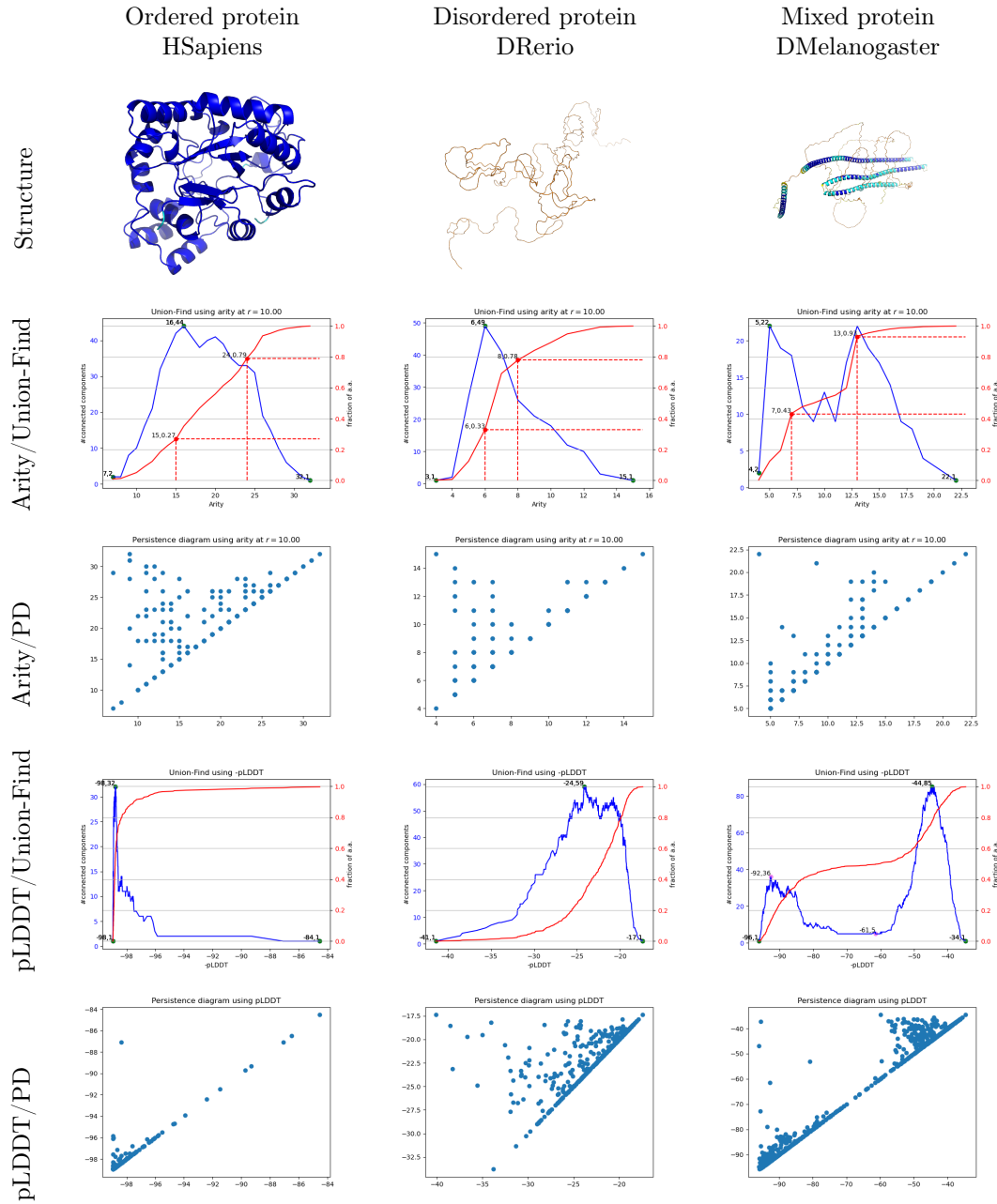


Figure 3: Filtrations $\mathcal{G}_{\text{arity}}$ and $\mathcal{G}_{\text{pLDDT}}$ on three prototypical examples corresponding to ordered, disordered and mixed proteins: evolution of the number of connected components and persistence diagrams. (Left) (H. Sapiens; AF-P15121-F1-model_v4; 316 amino acids; $H_p \sim 0.04$.) (Middle) (Zebrafish; AF-A0A0G2L439-F1-model_v4; 449 amino acids; $H_p \sim 0.27$.) (Right) (D. Melanogaster; AF-Q9VQS4-F1-model_v4; 781 amino acids; $H_p \sim 0.28$.) Second row, $\mathcal{G}_{\text{arity}}$: the blue curve codes the variation of the number of connected components; the red one is the cumulated fraction of C_α carbons whose arity is less than a threshold. Third row, $\mathcal{G}_{\text{arity}}$: persistence diagram for the connected components. Fourth row, $\mathcal{G}_{\text{pLDDT}}$: same as the second row, using pLDDT instead of arity. Fifth row, $\mathcal{G}_{\text{pLDDT}}$: same as the third row, using pLDDT instead of arity.

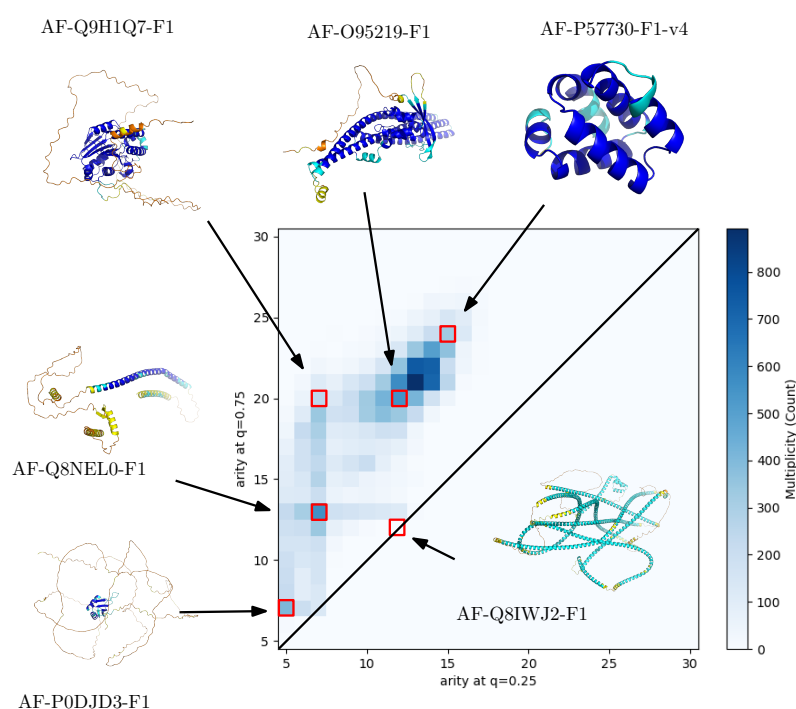


Figure 4: **H. Sapiens: arity map discriminates well-folded, intrinsically disordered, and low-confidence structures.** Upper triangle of the arity map: protein count with for a prescribed arity signature. Each bin corresponds to all AlphaFold predictions harboring a given arity signature.

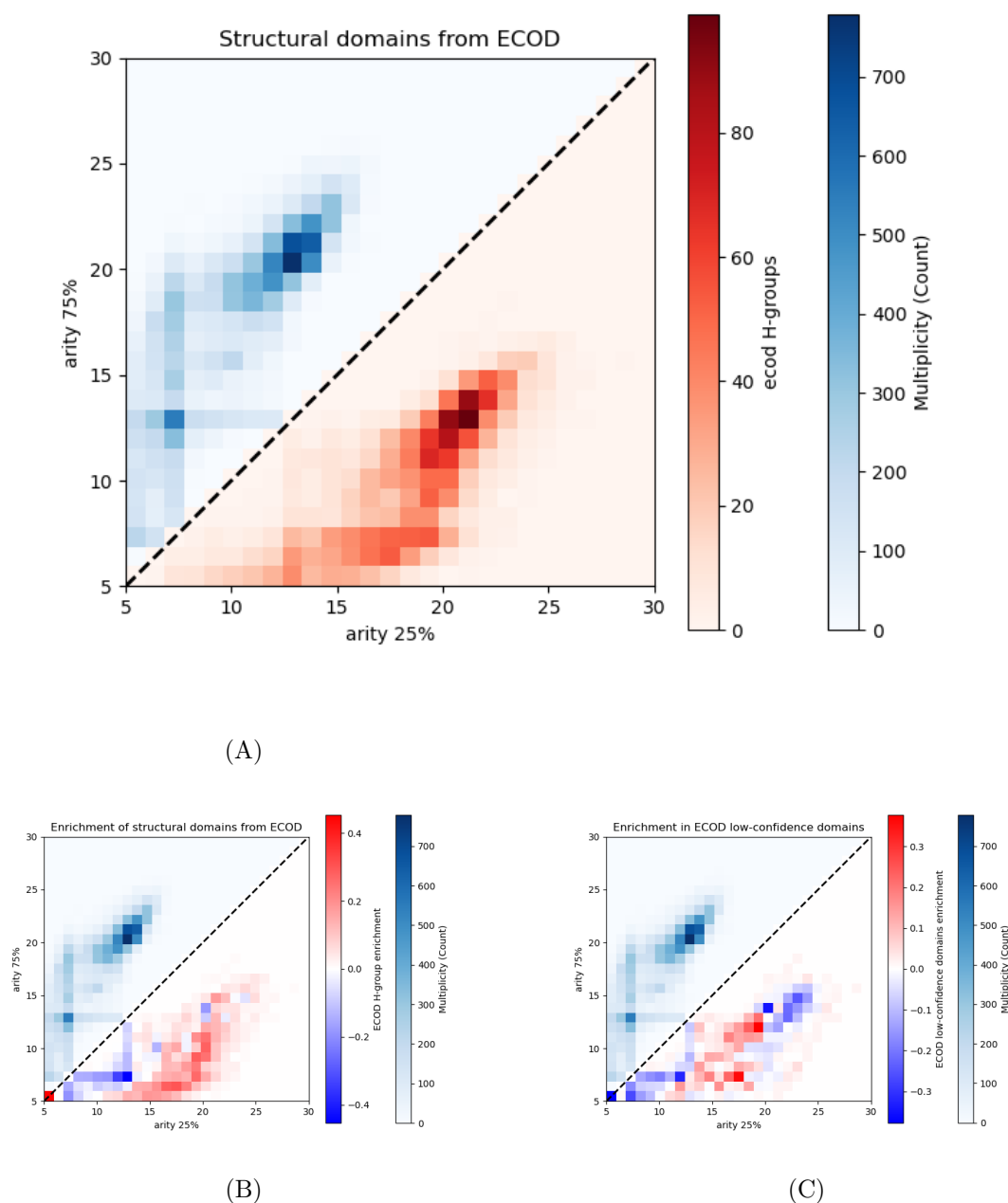


Figure 5: **H. Sapiens: arity map of the human proteome vs ECOD domain enrichment in human proteins.** The upper triangle always reports the human proteome arity map for reference. **(A)** Number of unique ECOD H-groups for each arity signature. **(B)** Enrichment difference (with respect to the proteome) of unique ECOD H-groups. **(C)** Enrichment difference (with respect to the proteome) of non-redundant ECOD low-confidence domains.

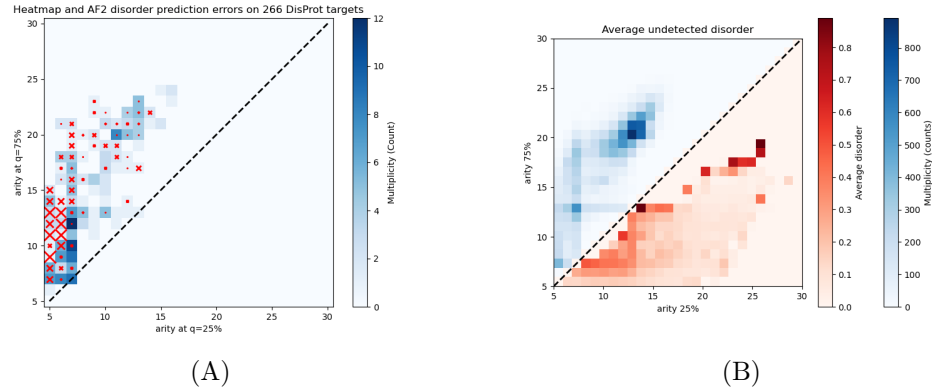


Figure 6: **H. Sapiens: arity map discriminates IDRs.** (A) DisProt analysis on 266 target proteins lacking structural data. The size of the red crosses over the arity map indicates the fraction of amino acids incorrectly predicted as disordered—pLDDT < 50% but lacking a disorder annotation in DisProt. (No cross: never, pixel-size cross: more than 70% of the times). The leftmost region of the plot shows a high enrichment in such cases. (B) AIUPred analysis. The upper triangle reports the human proteome arity for reference.

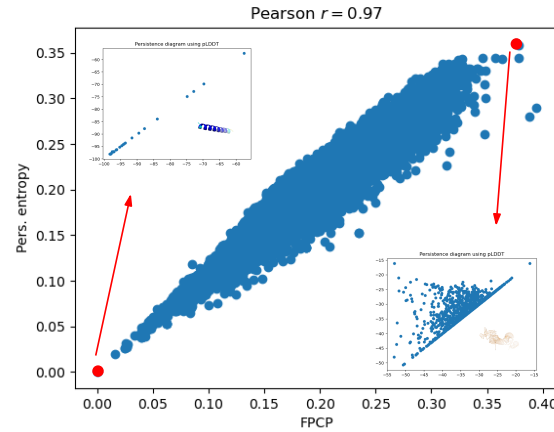


Figure 7: The persistence entropy H_p and the fraction of positive critical points (FPCP) f_{cp}^+ of $\mathcal{G}_{pLDDT}$ show a continuum between highly coherent and random-like structures in terms of pLDDT. Illustrations for HSapiens, with insets displaying the structures achieving the minimum and maximum H_p values.

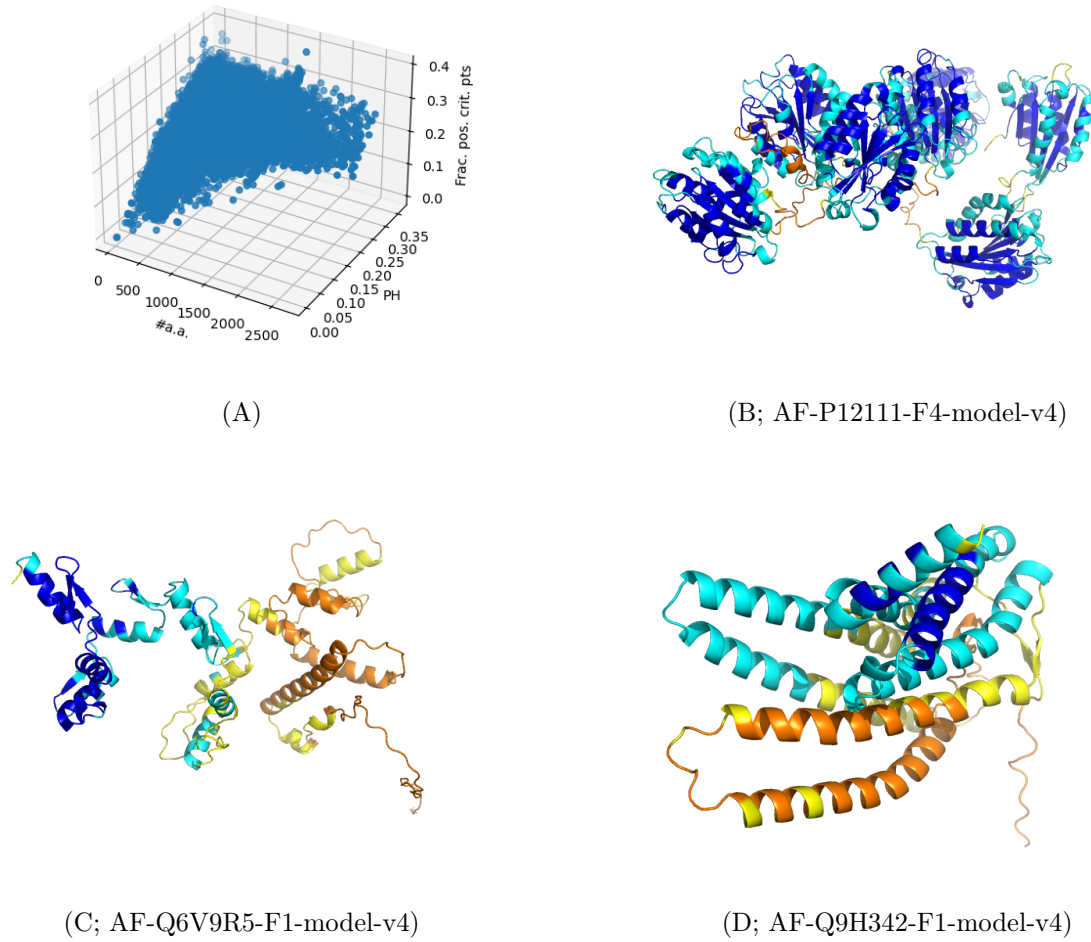


Figure 8: **Illustration of model fragmentation: wide range of pLDDT values aggregated in coherent stretches of pLDDT values along the sequence.** (A) Scatter plot with protein size $\times$ persistence entropy $H_p \times$ fraction of positive critical points f_{cp}^+ . (B, C, D) For HSapiens, at persistence threshold $t_p = 0.025$, 86 structures are characterized by $H_p \geq 0.25$, $\#a.a. \geq 200$, $PLM \geq 3$. Three of them are displayed.